Type MS · Normal, boundary and erroneous test data

Answer 1 of 6 · Question 2 part c.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c.i in this mark scheme extract

2(a)	A1	**The only correct answer is D** ***A*** *is not correct because abstraction is a technique used in computational thinking* ***B*** *is not correct because encryption is a security measure* ***C*** *is not correct because iteration is a programming construct*		**1**

Question	mp	Answer	Additional guidance	Mark
2(b)(i)		Award **one** mark for each correct cell:		
	B1 B2	real/double/float: (1) • pay rate/18.75 (1) • total pay/356.25 (1) • tax/0.2/71.25 (1) • income/285.00 (1) int/integer: (1) • hours/19 (1)	Example must match data type for B2 Award suitable example even if not taken from payslip Accept string and matching example e.g. Hours	**2**

5 of 23

Type MS · Normal, boundary and erroneous test data

Answer 1 of 6 · Question 2 part c.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
2(b)(ii)		Award **one** mark for each correct cell:		
	B3	**Input:** • Pay Rate/18.75 (1) • Hours/19 (1)		
	B4	**Process**: • Pay Rate * Hours/18.75 * 19 (1) • Total Pay * Tax/356.25 * 0.2 (1) • Total Pay – Tax/356.25 – 71.25 (1)	Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours: Total pay = pay rate * hours Accept 0.2 or 20% as part of calculation for tax Ok to mix types e.g. Total Pay * 20% Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)*	
	B5	**Output:** • Total Pay/356.25 (1) • Tax/71.25 (1) • Income/285 (1)		**3**

6 of 23

Type MS · Normal, boundary and erroneous test data

Answer 1 of 6 · Question 2 part c.i · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
2(c)(i)	C1 C2	Award **one** mark for **one** correct cell: Award **two** marks for all **three** correct cells: • **Normal:** any value between 0-255 e.g. 125 • **Boundary:** 0/255 • **Erroneous:** any negative value/any value 256 or above	Accept -1/256 for boundary data Don't accept non-int types	**2**

Question	mp	Answer	Additional Guidance	Mark
2(c)(ii)		Award **one** mark for any of the following up to a maximum of **four** marks:		
	C3	Input changed to int before comparison (1)		**4**
	C4	Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1)	Allow: >0 and/or <255	
	C5	'and' logical operator used to combine relational tests (1)	or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable	
	C6	Calling of `getHex(userInput)` (1)	Call must be inside print ( ) Subprogram and argument must be functionally correct Ignore additional variables/strings in print statement	

7 of 23

Type MS · Normal, boundary and erroneous test data

Answer 1 of 6 · Question 2 part c.i · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# complete the line to convert the user input to integer
userInput = int(input("please enter a number between 0 and 255 \n"))

# complete the test to check the input is between 0 and 255
if (userInput >=0) and (userInput <256):
    # complete the line to call getHex subprogram
    print(getHex(userInput))
else:
    print("number out of range")
```

8 of 23

Type MS · Normal, boundary and erroneous test data

Answer 1 of 6 · Question 2 part c.i · 2 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 1 of 6 · find part c in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
2 (c)		Award 1 mark for each set of test data.		

		Test data	Expected results
C1 C2 C3	booksSold profit	Either of • booksSold = 4 • profit = 4	Poor performances this week
	booksSold	5	Sales and profit are good this week
	profit	10	
	booksSold	21	Sales and profit are excellent this week
	profit	20	

(3)

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 2 of 6 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 3 of 6 · find part c in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
2 (a)	Award one mark for each of:		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the assignment of username and/or password Do not penalise spelling mistakes and alternative wording of the output.	(11)
	A1	At least one variable with a suitable variable name		
	A2	username = bard423		
	A3	password = nX2934?		
	A4	Loop used		
	A5	Username or password entered		
	A6	Username or password stored in variable(s)		
	A7	At least one suitable input message		
	A8	Checks username and password		
	A9	Appropriate error message(s) displays		
	A10	Welcome message displayed		
	A11	Executing and producing correct output (must have A4)		
Code example				
Python				

```
# Initialise variables
username = "bard423"
password = "nX2934?"
count = 0
inputUsername = ""
inputPassword = ""

# Print promts, take and check input from user
while inputUsername != username or inputPassword != password:
    if count > 0:
        print("There is a problem with the login details. Try again")
    count = count + 1
    inputUsername = input("Enter your username ")
    inputPassword = input("Enter your password ")
print("Welcome")
```

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 4 of 6 · find part c in this mark scheme extract

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 5 of 6 · find part c in this mark scheme extract

Question	Mp	Answer		Additional Guidance	Mark
2 (b)	Award 1 mark for each correct condition.			Accept alternatives e.g. Line 11 booksSold <=4 etc.	(4)
		Condition	**Output message**		
	B1	Number of books sold is under 5 or profit made is under 5	Poor performance this week		
	B2	Number of books sold is over 20; profit made is at least 20	Sales and profit are excellent this week		
	B3	Number of books sold is at least 5; profit made is at least 10	Sales and profit are good this week		
	B4	All other inputs	Alert manager		
Code example					
Python					

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

Type MS · Normal, boundary and erroneous test data

Answer 2 of 6 · Question 2 part c · 3 marks

2021 JUN · Paper 02 · extract 6 of 6 · find part c in this mark scheme extract

Question	ans	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 1 of 11 · find part a.vi in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 2 of 11 · find part a.vi in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 3 of 11 · find part a.vi in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 4 of 11 · find part a.vi in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 5 of 11 · find part a.vi in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 6 of 11 · find part a.vi in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 7 of 11 · find part a.vi in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

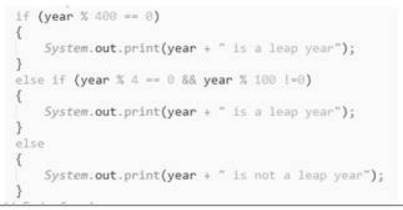
2021 NOV · Paper 02 · extract 8 of 11 · find part a.vi in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 9 of 11 · find part a.vi in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
 {
     Console.WriteLine(year + " is a leap year");
 }
else if (year % 4 == 0 && year % 100 !=0)
 {
     Console.WriteLine(year + " is a leap year");
 }
 else
 {
     Console.WriteLine(year + " is not a leap year");
 }
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 10 of 11 · find part a.vi in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Normal, boundary and erroneous test data

Answer 3 of 6 · Question 3 part a.vi · 2 marks

2021 NOV · Paper 02 · extract 11 of 11 · find part a.vi in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 1 of 9 · find part d in this mark scheme extract

1 (a)	A1	**The only correct answer is D** A is not correct because the value can be used more than once B is not correct because the value does not always have to be input C is not correct because the value does not always have to be used in a calculation		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 2 of 9 · find part d in this mark scheme extract

1 (b)	B1	**The only correct answer is B** A is not correct because this would not aid readability of the code C is not correct because this would not aid readability of the code D is not correct because this would not aid readability of the code		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 3 of 9 · find part d in this mark scheme extract

<table>
<tr><td>1 (c)</td><td>C1
C2</td><td>Award 1 mark for each of:
<table><tr><th>Data type</th><th>Example value</th></tr><tr><td>char</td><td>An example of any single letter, number or symbol</td></tr><tr><td>real</td><td>An example of a number with a decimal point</td></tr></table></td><td>Accept quotes around the character

Real number does not need a number after the decimal point to be awarded the mark</td><td>(2)</td></tr>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
</table>

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 4 of 9 · find part d in this mark scheme extract

1 (d)	D1	**Award up to 2 marks for a description such as:** • Boundary testing uses the most extreme valid data / minimum and maximum valid values (1) whereas erroneous testing uses invalid data (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 5 of 9 · find part d in this mark scheme extract

1 (e)	E1	Logic/logical		(1)

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

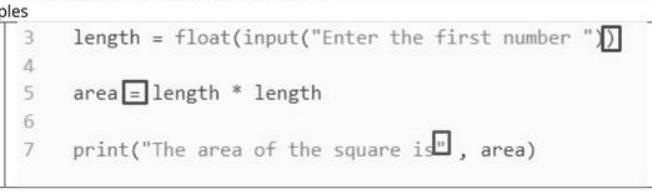
2022 JUN · Paper 02 · extract 6 of 9 · find part d in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 7 of 9 · find part d in this mark scheme extract

1 (f)	**Award 1 mark for each of:**			
	F1	Bracket added and no other amendments made to the line (1)		
	F2	== becomes = (1)		
	F3	" moved to after **is** and before , OR , area changed to +str(area) (1) Must still include the print statement		(3)
Code examples				
Python				

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 8 of 9 · find part d in this mark scheme extract

1 (g)	**Award 1 mark for each of:**			
	G1	At least 1 condition correct (1)		
	G2	At least 1 > or < condition and message match (1)		
	G3	At least 1 output includes letter and stored letter with an appropriate message (1)		
	G4	All conditions and outputs correct (1)		(4)
Code examples				
Python				

```
# Amend the code by completing the if statement
if letter > storedLetter:
    print(letter + " is later in the alphabet than " + storedLetter)
elif letter == storedLetter:
    print(letter + " is the same as " + storedLetter)
else:
    print(letter + " is earlier in the alphabet than " + storedLetter)
```

Type MS · Normal, boundary and erroneous test data

Answer 4 of 6 · Question 1 part d · 2 marks

2022 JUN · Paper 02 · extract 9 of 9 · find part d in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 5 of 6 · Question 2 part b.i · 3 marks

2023 JUN · Paper 02 · extract 1 of 4 · find part b.i in this mark scheme extract

2(a)(i)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1) • May include specialised/proprietary functions (1) • Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1) • making it more readable / makes code easier to understand (Code files become shorter)	Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets Don't allow quicker/easier/better without a reason being given Accept answers about subprograms as candidate may have written these themselves Line numbers are for guidance – mark answer as a whole	**2**
2(a)(ii)	A3	**The only correct answer is A** ***B*** *is not correct because iteration is not callable and does not return a result.* ***C*** *is not correct because a procedure, while callable, does not return a result.* ***D*** *is not correct because selection is not callable and does not return a result.*		**1**
2(a)(iii)	A4	Award **one** mark for: • A variable that exists/is accessible only in the subprogram/scope in which it is created (1)		**1**

Type MS · Normal, boundary and erroneous test data

Answer 5 of 6 · Question 2 part b.i · 3 marks

2023 JUN · Paper 02 · extract 2 of 4 · find part b.i in this mark scheme extract

Type MS · Normal, boundary and erroneous test data

Answer 5 of 6 · Question 2 part b.i · 3 marks

2023 JUN · Paper 02 · extract 3 of 4 · find part b.i in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(b)(i)</td><td>B1
B2
B3</td><td>Award one mark for each correct row.
<table><tr><th></th><th>Adult</th><th>Children</th></tr><tr><td>Normal</td><td>1/2/3/4</td><td>1/2/3/4/5/6</td></tr><tr><td>Boundary</td><td>1/4</td><td>1/6</td></tr><tr><td>Erroneous</td><td>Any value <1 or >4</td><td>Any value <1 or >6</td></tr></table></td><td>Accept all valid values for normal and boundary
Do not accept written numbers (two) as question asks for ‘numeric data’ – this is testing range, not data types.</td><td>3</td></tr>
<tr><td rowspan="5">2(b)(ii)</td><td colspan="2">Award one mark for each of:</td><td rowspan="5">Runtime error (B7) could be fixed before logic error (B6)
Applies to all languages
Accept < for <=
Allow /-1000 as alternative to B6
/1000 could happen after initial location e.g. print(“Order”, weightNeed/1000, “kilograms)</td><td rowspan="5">4</td></tr>
<tr><td>B4</td><td>Change =+= to +</td></tr>
<tr><td>B5</td><td>Change > to <=</td></tr>
<tr><td>B6</td><td>Change order of subtraction to: requiredWeight - inStock</td></tr>
<tr><td>B7</td><td>Change 0 to 1000</td></tr>
</table>

Type MS · Normal, boundary and erroneous test data

Answer 5 of 6 · Question 2 part b.i · 3 marks

2023 JUN · Paper 02 · extract 4 of 4 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 1 of 14 · find part d in this mark scheme extract

2(a)(i)	A1	**The only correct answer is A** ***B*** *is not correct because* ***computational thinking*** *are skills used in creating algorithms and programs* ***C*** *is not correct because* ***decomposition*** *is breaking down a problem into smaller parts* ***D*** *is not correct because* ***pseudocode*** *is a notation for expressing an algorithm*		**1**

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 2 of 14 · find part d in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 3 of 14 · find part d in this mark scheme extract

2(a)(ii)	A2	Award **one** mark for: • Variable (1)	Allow parameter Ignore local/global	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 4 of 14 · find part d in this mark scheme extract

2(b)(i)	B1	Award **one** mark for any of the following: • An instruction is impossible to execute (1) • The program stops executing **unexpectedly** / **crashes** (1)	Do not award an example, as that is the next question Don't award repeat of question e.g. "Stopping during execution" without clarifying reason or that stop was unexpected	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 5 of 14 · find part d in this mark scheme extract

2(b)(ii)	B2	Award **one** mark for any of the following: • Division by zero (1) • Accessing memory outside the range of an array (1) • Attempting an operation on incompatible types (1) • File not found (1) • An exception is raised that cannot be/is not handled (1)	Award other examples, if they can be mapped to a bullet	**1**

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 6 of 14 · find part d in this mark scheme extract

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 7 of 14 · find part d in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 8 of 14 · find part d in this mark scheme extract

2(c)		Award **one** mark for each of:		
	C1	Add or remove a semicolon on initialisation of the length variable (1)	• `int length = 4;` • `int length = 4;` • `length = 4`	
	C2	Add a comma between the arguments to the subprogram call to generate a random number (1)	• `width = randGenerator.Next (1, 6);` • `width = randGenerator.nextInt(1, 6);` • `width = random.randint (1, 5)`	
	C3	Replace modulus operator (%) with multiplication operator (*) (1)	• `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length)`	**3**

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 9 of 14 · find part d in this mark scheme extract

C#

```
// Q02cFINISHED

int width = 0;
int length = 4;
int perimeter = 0;
Random randGenerator = new Random();

width = randGenerator.Next (1, 6);
perimeter = (2 * width) + (2 * length);

Console.WriteLine (width.ToString() + " " +
                   length.ToString() + " " +
                   perimeter.ToString());
```

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 10 of 14 · find part d in this mark scheme extract

Java

```java
// Q02cFINISHED

import java.util.Random;

public class Q02cFINISHED
{
    public static void main (String[] args)
    {
        int width = 0;
        int length = 4;
        int perimeter = 0;
        Random randGenerator = new Random();

        width = randGenerator.nextInt(1, 6);
        perimeter = (2 * width) + (2 * length);

        System.out.println (String.valueOf(width) + " " +
                            String.valueOf(length) + " " +
                            String.valueOf(perimeter));
    }
}
```

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 11 of 14 · find part d in this mark scheme extract

Python

```python
# Q02cFINISHED

import random

width = 0
length = 4
perimeter = 0

width = random.randint (1, 5)
perimeter = (2 * width) + (2 * length)

print (width, length, perimeter)

```

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 12 of 14 · find part d in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 13 of 14 · find part d in this mark scheme extract

<table>
<tr>
<td>2(d)</td>
<td>D1
D2
D3
D4</td>
<td>
<table>
<tr><td>Erroneous</td><td>Award one mark for any of the following up to a maximum of two marks:
Contains incorrect data type (1)
<2 letters (1)
9+ letters (1)</td></tr>
<tr><td>Normal</td><td>A string between 2 and 8 in length that includes only letters (1)</td></tr>
<tr><td>Boundary</td><td>Award one mark for any of the following:
A string of exactly 8 letters (1)
A string of exactly 2 letters (1)</td></tr>
</table>
</td>
<td>Award any correct example (not description)
Do not award an empty cell as equivalent to empty string
Do not accept same test twice for erroneous</td>
<td>4</td>
</tr>
</table>

Type MS · Normal, boundary and erroneous test data

Answer 6 of 6 · Question 2 part d · 4 marks

2024 JUN · Paper 02 · extract 14 of 14 · find part d in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark